

Gayathri Puram

AI/ML Engineer

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SUMMARY

- Results-oriented AI/ML Engineer with over 3+ years of experience designing, developing, and deploying machine learning models and artificial intelligence solutions.
- Proven track record of delivering impactful projects, strong expertise in data preprocessing, feature engineering, model selection, and hyperparameter tuning.
- Skilled in Python programming and proficient in popular ML frameworks.
- Excellent problem-solving skills and ability to work collaboratively on complex projects.
- Proficiency in statistics, mathematical models, and AWS tools.

EDUCATION

Master of Science in Computer Science Arizona State University, Tempe, AZ, USA	Dec 2023
Bachelor of Science in Computer Science Amrita Vishwa Vidyapeetham, Coimbatore, Tamil Nadu	Jun 2021

TECHNICAL SKILLS

Methodology: SDLC, Agile, Waterfall

Languages: Python, C, C++, Java, R, SQL, MATLAB, HTML, CSS, JavaScript, TypeScript

Frameworks: TensorFlow, PyTorch, Keras, NumPy, Pandas, Scikit-Learn, Matplotlib

IDEs: Visual Studio Code, PyCharm, Jupyter Notebook

Machine Learning: Linear-logistic regression, Clustering, SVM, PCA, Random Forest, Boosting, Lasso, Ridge, CNN, RNN

Deep Learning: TensorFlow, Keras

AWS Services: Amazon SageMaker, Bedrock (retrieveAndGenerate, retrieve), AWS Comprehend, Amazon Polly, AWS Transcribe, AWS Rekognition, S3, Glue, Redshift, RDS, EC2, Lambda, AWS IAM, AWS Shield, AWS CloudFormation, Terraform, AWS CloudWatch, AWS X-Ray, AWS IAM, AWS CloudFormation, Terraform

Data Management: S3, Glue, Amazon Redshift, RDS

Compute Services: Amazon EC2, AWS Lambda

General Cloud and DevOps Skills: Cloud Architecture, CI/CD with CodePipeline, CodeBuild, CodeDeploy, Monitoring with CloudWatch, AWS X-Ray

Version Control Tools: Git, GitHub, GitLab

Operating Systems: Windows, Linux, Mac

Soft Skills: Problem-Solving, Communication, Collaboration

WORK EXPERIENCE

Fisglobal, USA AI/ML Engineer (SME)	Sep 2023 - Current
- Managed software projects, consistently using Agile methodologies to meet project deadlines and enhance client satisfaction. - Developed and trained an AI/ML model in AWS SageMaker, incorporating Langchain Chains and Agents, Gen AI, and prompt engineering to schedule appointments based on the FIFO principle, improving the efficiency of the scheduling process. - Utilized open-source routing API and Langchain's API interaction capabilities to sort solutions by frequency by matching feedback to closed tickets and sorting algorithms to segregate into different customer buckets. - Collected data for air passengers overstaying in the USA. Preprocessed 50,000 lines of data using data cleaning, and null elimination method in SQL database. - Led the CBP AI use case to predict the probability of air passengers' likelihood of overstaying their USA visit using AWS SageMaker, Redshift, Python, and Langchain's Chains and Agents. This model probability is planned to be utilized by NTC (National Targeting Center) in the future roadmap. - Created machine learning modules from the ground up on AWS SageMaker, researched the problem area, built a natural language processing model for semantic similarity, clustering, and summarizing text, deploying it on Amazon Web Services as Docker container and on GCP using Kubernetes. - Mentored Machine Learning and Physics interns (MIT graduates), utilizing agile methodology scheduling CI unit tests on Linux/Unix environment while sharing my knowledge in AWS SageMaker, Redshift, SQL, Python, Pytorch, and cloud computing architectures.	
Metasystems, India AI/ML Engineer	Jun 2019 - Nov 2021
- Led the development of a customer churn prediction model using XGBoost, reducing churn rate by 15% and increasing customer retention. - Designed and implemented a natural language processing (NLP) solution for sentiment analysis, improving customer feedback analysis by 20%. - Developed and deployed deep learning models, including Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), for image and sequence data, enhancing model performance by 30%. - Collaborated with cross-functional teams to integrate AI/ML solutions into the company's products, resulting in a 30% increase in user engagement. - Conducted data preprocessing, feature engineering, and model selection to optimize model performance, achieving accuracy rates above 90% on various projects. - Actively participated in the evaluation and selection of ML frameworks, contributing to the adoption of TensorFlow for deep learning projects. - Utilized my web development skills in HTML, CSS, and JavaScript to successfully deploy the model on a webpage using Flask API. - Collaborated with data scientists and engineers to deploy scalable machine learning solutions on AWS, utilizing services like SageMaker, EC2, and S3.	